



Jianke Yu 俞鉴珂, PhD

PhD (2026) in Graph Machine Learning & Databases

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RESEARCH INTERESTS

Machine learning for graph-structured data and database systems: graph neural network architectures for incomplete, heterogeneous and temporal graphs (fraud detection, malware detection, graph similarity computation), and learning-based methods that accelerate database and data-mining algorithms.

Keywords: Graph Neural Networks, Graph Databases, AI for Databases, Fraud Detection, Knowledge Graphs

EDUCATION

PhD, Information Technology · University of Technology Sydney 2023.07 – 2026.09
Conferred 9 September 2026. Supervised by Prof. Ying Zhang and A/Prof. Lu Qin. Thesis: "Fraud detection on the customer-product bipartite graph" (learning-based algorithms for graph-structured data and database systems). Published in KDD 2023 and IEEE TKDE (2024, 2025, 2026). Recipient of the UTS FEIT HDR Excellence Award, Individual Category, Distinction (2025).

M.Eng, Electronic Information · Zhejiang Gongshang University 2020.09 – 2023.06
Supervised by Prof. Xiaoyang Wang. Research on graph representation learning, fraud detection and graph similarity computation.

B.Eng, Electronic Information Engineering · Zhejiang Gongshang University 2016.09 – 2020.06

APPOINTMENTS & EXPERIENCE

Incoming Faculty Member · Zhejiang Gongshang University 2027.03 (incoming)
Joining as a faculty member under the university's talent introduction programme.

Research Intern · Token Foundry, Alibaba Token Hub, Alibaba Group 2025.12 – Present
Research on graph learning methods for industrial data systems.

Research Assistant · University of Technology Sydney 2023.07 – Present
ARC Discovery Project. Graph neural networks, graph similarity search and learning-based database algorithms.

Guest Lecturer · University of Technology Sydney 2025.01 – 2025.12
Guest lecturer for UTS 43023 Emerging Topics in AI (Spring 2025).

Research Intern · DAMO Academy / Alibaba Cloud, Alibaba Group 2022.07 – 2023.04
Fraud detection on e-commerce platforms (work later published in KDD 2023 and IEEE TKDE).

Research Intern · Taobao, Alibaba Group 2021.11 – 2022.03
Graph-based user modelling and risk analysis.

PUBLICATIONS

19 peer-reviewed papers (10 journal, 9 conference), 6 as first author, 7 CCF-A, 119 citations.

- [1] MGDN: A Graph of Graphs Neural Network for Malware Detection. *IEEE Transactions on Knowledge and Data Engineering*, 2026. **1st author** (CCF-A) (CAS Zone 1 Top) (JCR Q1)
- [2] SARL: A Scalable Attribute-Informed Representation Learning Framework for Bipartite Graphs. *IEEE Transactions on Knowledge and Data Engineering*, 2026. **1st author** (CCF-A) (CAS Zone 1 Top) (JCR Q1)
- [3] Temporal Insights for Group-Based Fraud Detection on e-Commerce Platforms. *IEEE Transactions on Knowledge and Data Engineering*, 2024. **1st author** (CCF-A) (CAS Zone 1 Top) (JCR Q1)
- [4] Group-based Fraud Detection Network on e-Commerce Platforms. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023. **1st author** (CCF-A)
- [5] Mining Discriminative Salient Objects with Optimal Transport for Few-Shot Image Classification. *FLINS-ISKE*, 2026.
- [6] Finding critical users in social networks with reinforcement learning. *Information Sciences*, 2026. (CCF-B) (CAS Zone 2) (JCR Q1)
- [7] AEFA: An Ensemble Framework for Fraud Detection in the Forex Market. *International Conference on Advanced Data Mining and Applications*, 2025. (CCF-C)
- [8] Deep Overlapping Community Search via Subspace Embedding. *ACM SIGMOD Conference*, 2025. (CCF-A)
- [9] RPDN: An Effective Rating Pollution Attacks Detection Framework for Recommendation Systems. *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, 2025. (CCF-C)

- [10] IGFM: An Enhanced Graph Similarity Computation Method with Fine-Grained Analysis. *Data Science and Engineering*, 2025. (CCF-B) (CAS Zone 1 Top) (JCR Q2)
- [11] RIDA: a robust attack framework on incomplete graphs. *World Wide Web*, 2025. 1st author (CCF-B) (CAS Zone 4) (JCR Q2)
- [12] Neural Similarity Search on Supergraph Containment (Extended Abstract). *IEEE International Conference on Data Engineering*, 2024. (CCF-A)
- [13] SBGMN: A Multi-view Sign Prediction Network for Bipartite Graphs. *Asia Pacific Web Conference (APWeb-WAIM)*, 2024. (CCF-C)
- [14] IFGNN: An Individual Fairness Awareness Model for Missing Sensitive Information Graphs. *Australasian Database Conference*, 2023.
- [15] KGNIE: A Learning Method for Estimating Node Importance in Knowledge Graphs. *IEEE International Conference on High Performance Computing and Communications*, 2023. (CCF-C)
- [16] FPGN: follower prediction framework for infectious disease prevention. *World Wide Web*, 2023. 1st author (CCF-B) (CAS Zone 4) (JCR Q2)
- [17] Neural Similarity Search on Supergraph Containment. *IEEE Transactions on Knowledge and Data Engineering*, 2023. (CCF-A) (CAS Zone 1 Top) (JCR Q1)
- [18] Bipartite graph capsule network. *World Wide Web*, 2022. (CCF-B) (CAS Zone 4) (JCR Q2)
- [19] Polarity-based graph neural network for sign prediction in signed bipartite graphs. *World Wide Web*, 2022. (CCF-B) (CAS Zone 4) (JCR Q2)

I INVITED TALKS

Scalable Big Data Computation · Seminar, School of Data Science, The Chinese University of Hong Kong, Shenzhen 18 Apr 2026

I HONORS & AWARDS

Award for Outstanding Overseas Study Elite (2025–2026) · China Scholarship Council 2026
National-level award for outstanding self-financed PhD students abroad.

UTS FEIT HDR Excellence Award — Individual Category, Distinction · University of Technology Sydney, Faculty of Engineering and IT 2025
Faculty-level award (Distinction) for research excellence during HDR candidature.

UTS ARC Discovery Project Scholarship (Full Scholarship) · University of Technology Sydney / Australian Research Council 2023

UTS International Research Scholarship · University of Technology Sydney 2023

Excellent Practical Achievement Award for Professional Master Students · Education Department of Zhejiang Province 2022
Provincial award for outstanding professional master's practice achievements.

Second Prize, 4th Zhejiang Province University Student Robotics Competition · Zhejiang Province 2019

Second Prize, "Internet+" Innovation and Entrepreneurship Competition · Zhejiang Gongshang University 2019

I ACADEMIC SERVICE

Peer Reviewer · IEEE Transactions on Knowledge and Data Engineering (TKDE) 2025

Platform Administrator · Research Computing Platform, Prof. Ying Zhang & Prof. Xiaoyang Wang's Team, Zhejiang Gongshang University 2024–

I TEACHING

Guest Lecturer · 43023 Emerging Topics in Artificial Intelligence, University of Technology Sydney Spring 2025

Teaching Assistant · COMP9311 Database Systems, UNSW Sydney Term 3 2023

Teaching Assistant · DATA1001 Introduction to Data Science, UNSW Sydney Term 2 2024, Term 2 2025

I SKILLS & LANGUAGES

Technical Skills Graph Neural Networks, Python / PyTorch, Graph Databases & Query Processing, Data Mining & Fraud Detection

Research & Service Academic Writing & Publishing, Peer Review (IEEE TKDE), Teaching & Student Supervision

Languages Chinese (Native), English (Fluent)